Equal Equilibria and Different Relaxation Spectra: A Proof of Conjecture 00000006640

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Abstract

Two symmetric two-state master equations, with jump rates one and two, have the same static solution space and the same unique stationary probability vector. Their generator spectra are $\{0, -2\}$ and $\{0, -4\}$, and their propagation spectra differ at every positive time. The construction, stochastic semigroups, differential equations, exact spectra, and existential claim are verified in Lean using actual real matrices and derivatives.

Reading of the source. Both language versions ask for an explicit pair of master equations with identical static solution sets but different time spectra. For a finite-state continuous-time master equation, write probability vectors as columns and use

$$p'(t) = Qp(t), \qquad p_i(t) \geq 0, \qquad \sum_i p_i(t) = 1.$$

The generator has nonnegative off-diagonal entries and zero column sums. Static probability solutions are $\{p : Qp = 0,\ p_i \geq 0,\ \sum_i p_i = 1\}$; without probability normalization the static solutions form $\ker Q$. We establish equality in both senses. We establish different spectra both for the generator and for the propagator at every fixed $t > 0$. The generators and propagators are real symmetric and obey detailed balance for the same positive probability vector, so the example also has the usual time-reversal symmetry. Time is measured in the same units in both systems; the source imposes no quotient by time rescaling.

Theorem 1. *For $r > 0$ define the two-state generator and probability vector*

$$Q_r = \begin{pmatrix} -r & r \\ r & -r \end{pmatrix}, \qquad \pi = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}.$$

The systems $p' = Q_1 p$ and $p' = Q_2 p$ satisfy:

1. *They are valid reversible master equations with the common unique stationary probability vector π, and $\ker Q_1 = \ker Q_2 = \{(a, a)^{\mathsf{T}} : a \in \mathbb{R}\}$.*
2. *Their generator spectra are $\{0, -2\}$ and $\{0, -4\}$.*
3. *Their stochastic fundamental solutions are*

$$P_r(t) = \frac{1}{2} \begin{pmatrix} 1 + e^{-2rt} & 1 - e^{-2rt} \\ 1 - e^{-2rt} & 1 + e^{-2rt} \end{pmatrix}, \tag{1}$$

   with $P_r(0) = I$, $P_r(s + t) = P_r(s)P_r(t)$, and $P_r'(t) = Q_r P_r(t)$. For every initial probability vector p_0, $p(t) = P_r(t)p_0$ remains a probability vector for $t \geq 0$ and solves the master equation with $p(0) = p_0$.
4. *For each $t > 0$, the propagator spectra $\{1, e^{-2t}\}$ and $\{1, e^{-4t}\}$ differ.*

Thus the source's requested separation exists, including for symmetric reversible systems.

Proof. For $r > 0$ the off-diagonal generator entries equal r and both column sums vanish. The matrix is symmetric and $(Q_r)_{ij}\pi_j = (Q_r)_{ji}\pi_i$, proving detailed balance. For a column $p = (x, y)^\mathsf{T}$,

$$Q_r p = r(y - x, x - y)^\mathsf{T}.$$

It vanishes exactly when $x = y$. Imposing $x + y = 1$ yields $x = y = 1/2$, a nonnegative vector. This proves both static assertions.

The characteristic determinant is

$$\det(\mu I - Q_r) = (\mu + r)^2 - r^2 = \mu(\mu + 2r).$$

For a finite real matrix the real algebraic spectrum consists exactly of values for which this determinant vanishes, so $\sigma(Q_r) = \{0, -2r\}$. The two chosen spectra are distinct, since -2 belongs to the first but not the second. Equivalently, the constant mode $(1, 1)^\mathsf{T}$ has eigenvalue zero and the imbalance mode $(1, -1)^\mathsf{T}$ has eigenvalue $-2r$.

In (1), $0 < e^{-2rt} \le 1$ for $r, t \ge 0$, so the entries are nonnegative and the columns sum to one. Direct substitution gives $P_r(0) = I$. The identity $e^{-2r(s+t)} = e^{-2rs}e^{-2rt}$ gives the semigroup law by matrix multiplication. Differentiating the entries gives

$$P_r'(t) = re^{-2rt}\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} = Q_r P_r(t).$$

These identities hold for all real s, t; stochasticity is used only for nonnegative time. Multiplication by a fixed initial column gives $p' = Q_r p$ and the stated initial condition. Nonnegativity and column sums preserve the probability constraints. Also $P_r(t)\pi = \pi$.

Put $a = e^{-2rt}$. A second determinant calculation gives

$$\det(\mu I - P_r(t)) = (\mu - 1)(\mu - a), \qquad \sigma(P_r(t)) = \{1, e^{-2rt}\}.$$

For $t > 0$, strict monotonicity of the real exponential gives $0 < e^{-4t} < e^{-2t} < 1$. Therefore e^{-2t} belongs to the first propagator spectrum but not the second. The two decay rates are different even though their static states coincide. □

Formal verification and scope

The file `lean/Main.lean` uses Lean 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

The state space is `Fin 2`; generators and propagators are actual 2×2 matrices over $\mathbb{R}$. The probability predicate includes both nonnegativity and normalization. The static predicate is defined using actual matrix-vector multiplication.

The formalization verifies the generator conditions, Hermitian symmetry, detailed balance, the unique normalized static vector, and equality of the full unnormalized kernels. The exact spectra use Mathlib's algebraic `spectrum`, not a list defined to be the claimed eigenvalues. They are obtained by reducing noninvertibility to the determinant and proving the determinant factorizations. The propagators satisfy the initial condition and semigroup law. They preserve nonnegativity and total probability. Their derivatives are actual `HasDerivAt` statements. For every initial vector, the trajectory satisfies the master equation in each coordinate. This finite-dimensional existence claim does not require a probability path construction.

The final theorem is

`Conjecture6640.conjecture_6640_true`.

It proves `SeparationClaim`: there exist two generators, two stochastic fundamental solutions, and a common detailed-balance probability vector; the normalized static sets and full kernels coincide, the generator spectra differ, and the propagator spectra differ at every positive time. The actual mathematical objects and the complete existential statement are thus present in Lean. The proof does not invoke a general matrix-exponential or ODE uniqueness theorem; it verifies the displayed explicit fundamental solutions directly. All mathematical assertions required by the theorem are formalized. The displayed eigenvector explanation is an additional paper explanation of the formally proved determinant calculation.

The audited main and bridge theorems use only `propext`, `Classical.choice`, and `Quot.sound`. This submission was developed by a delegated AI agent with a self-review; the coordinating agent reviews it separately. No external independent review is claimed.

Source and reproduction

The exact English and Chinese source appears in `SOURCE.md`, from upstream commit

`b355e42906e6eaf52bc7e9d005e0ef02d8aa7400`.

Its hash and retrieval information are in

`verification/SOURCE_PROVENANCE.md`.

The argument is elementary and self-contained. The standard Q-matrix and master-equation framework is discussed, for example, in J. R. Norris, *Markov Chains*, Chapter 2, author's book page. The column-vector convention used here is explicitly specified above.

With the pinned dependencies available, run in the `lean` directory:

```
lake env lean -DwarningAsError=true Main.lean
```

and `lake build` for a fresh build. Compile `main.tex` with Tectonic or a compatible LaTeX engine.